

English Version

Single Modern Design

EV vs ICE Vehicle Specs

Preprocessing & EDA Report

Dataset: EV_vs_ICE_Vehicle_Specs_2015_2026.csv

Purpose: To document a clean, reproducible Jupyter Notebook workflow for exploratory data analysis and preprocessing.

ROWS

15,301

COLUMNS

14

MISSING VALUES

1

DUPLICATES

0

1. Project Overview

Objective

This report summarizes the preprocessing and EDA steps for a vehicle specification dataset containing EV and ICE vehicles from 2015 to 2026. The workflow is designed for Jupyter Notebook and prepares the dataset for downstream machine learning tasks.

Recommended prediction target

Vehicle_Category is the most practical target for a first model. The model can learn to classify vehicles such as *EV*, *ICE (Gas)*, *ICE (Diesel)*, and *Hybrid/Other*.

Quick insights

- The dominant class is **ICE (Gas)** (88.81% of the dataset).
- Only **1** missing value is present, in **Fuel_Type**.
- The dataset spans from **2015** to **2026**.
- There are **8** numeric and **6** categorical features.

2. Dataset Structure

Columns and data types

Column	Data Type
Make	object
Model	object
Year	int64
Fuel_Type	object
Engine_Cylinders	float64
Engine_Size_L	float64

Missing values

Column	Missing Count
Fuel_Type	1

The dataset is already very clean. Only one missing value was found and it can safely be filled with **"Unknown"**.

Column	Data Type
Drivetrain	object
Transmission	object
City_MPG	int64
Highway_MPG	int64
Combined_MPG	int64
CO2_Emissions_g_per_mile	int64
EV_Range_miles	int64
Vehicle_Category	object

3. Jupyter Notebook Code Structure

IMPORTS

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
```

LOAD AND INSPECT THE DATASET

```
df = pd.read_csv("EV_vs_ICE_Vehicle_Specs_2015_2026.csv")

print(df.head())
print(df.shape)
print(df.info())
print(df.describe())
print(df.isnull().sum())
print(df.duplicated().sum())
```

CLEANING AND FEATURE SIMPLIFICATION

```
df_clean = df.copy()
df_clean["Fuel_Type"] = df_clean["Fuel_Type"].fillna("Unknown")
df_clean = df_clean.drop_duplicates()

def simplify_transmission(value):
    text = str(value).lower()
    if "manual" in text:
        return "Manual"
    if "automatic" in text:
        return "Automatic"
    return "Other"

df_clean["Transmission_Type"] = df_clean["Transmission"].apply(simplify_transmission)
df_clean = df_clean.drop(columns=["Transmission", "Model"])
print(df_clean.head())
```

EDA EXAMPLES

```
numeric_cols = df.select_dtypes(include=["int64", "float64"]).columns

plt.figure(figsize=(8, 5))
df["Vehicle_Category"].value_counts().plot(kind="bar")
plt.title("Vehicle Category Distribution")
plt.show()

plt.figure(figsize=(10, 7))
plt.imshow(df[numeric_cols].corr(), aspect="auto")
plt.colorbar()
plt.title("Correlation Matrix")
plt.show()
```

PREPROCESSING PIPELINE FOR MODELING

```
target = "Vehicle_Category"
X = df_clean.drop(columns=[target])
y = df_clean[target]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y,
)
```

```
numeric_features = X.select_dtypes(include=["int64", "float64"]).columns
categorical_features = X.select_dtypes(include="object").columns

numeric_transformer = Pipeline(steps=[
    ("scaler", StandardScaler())
])

categorical_transformer = Pipeline(steps=[
    ("encoder", OneHotEncoder(handle_unknown="ignore"))
])

preprocessor = ColumnTransformer(
    transformers=[
        ("num", numeric_transformer, numeric_features),
        ("cat", categorical_transformer, categorical_features),
    ]
)

X_train_processed = preprocessor.fit_transform(X_train)
X_test_processed = preprocessor.transform(X_test)
```

EXPORT CLEANED DATASET

```
df_clean.to_csv("cleaned_EV_vs_ICE_vehicle_specs.csv", index=False)
```

4. Exploratory Data Analysis

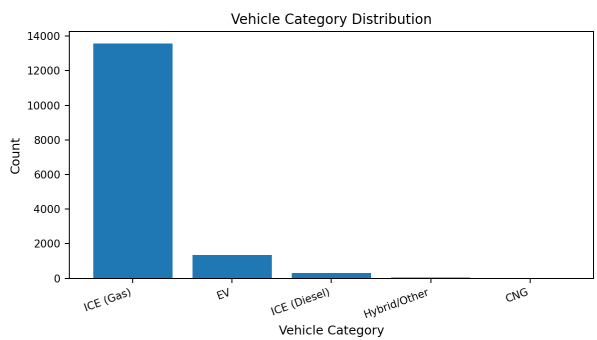


Figure 1. Distribution of vehicle categories in the dataset.

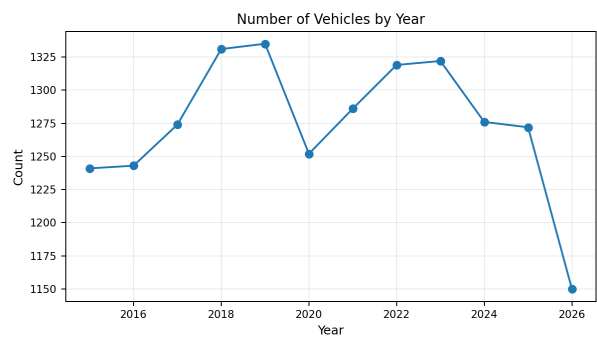


Figure 2. Number of records by year.

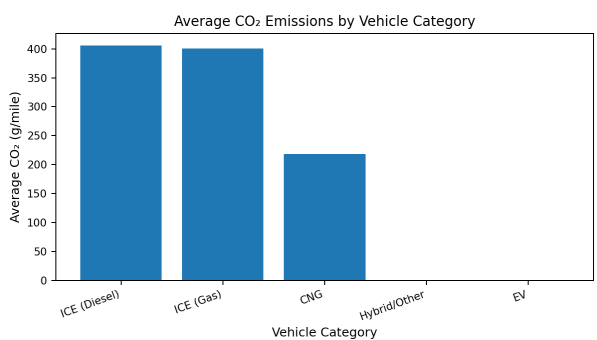


Figure 3. Average CO₂ emissions by vehicle category.

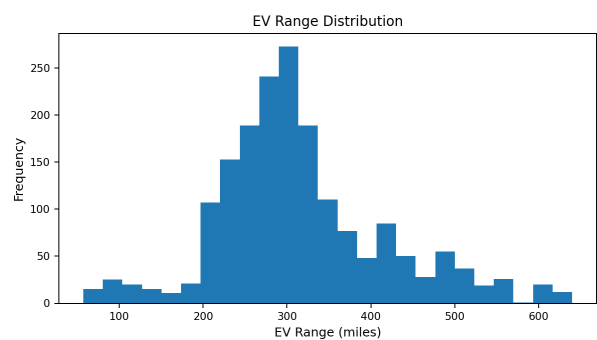


Figure 4. Distribution of EV range values.

EDA interpretation

- **Class imbalance:** ICE (Gas) vehicles are heavily dominant, so classification models may require stratified splitting and possibly imbalance handling.
- **Temporal coverage:** The dataset covers multiple years, which makes time-based comparisons possible.
- **Environmental separation:** CO₂ emissions clearly separate EVs from combustion-based vehicles.
- **EV-specific behavior:** EV range is a meaningful feature only for electric vehicles, so interpretation should consider category context.

Correlation matrix

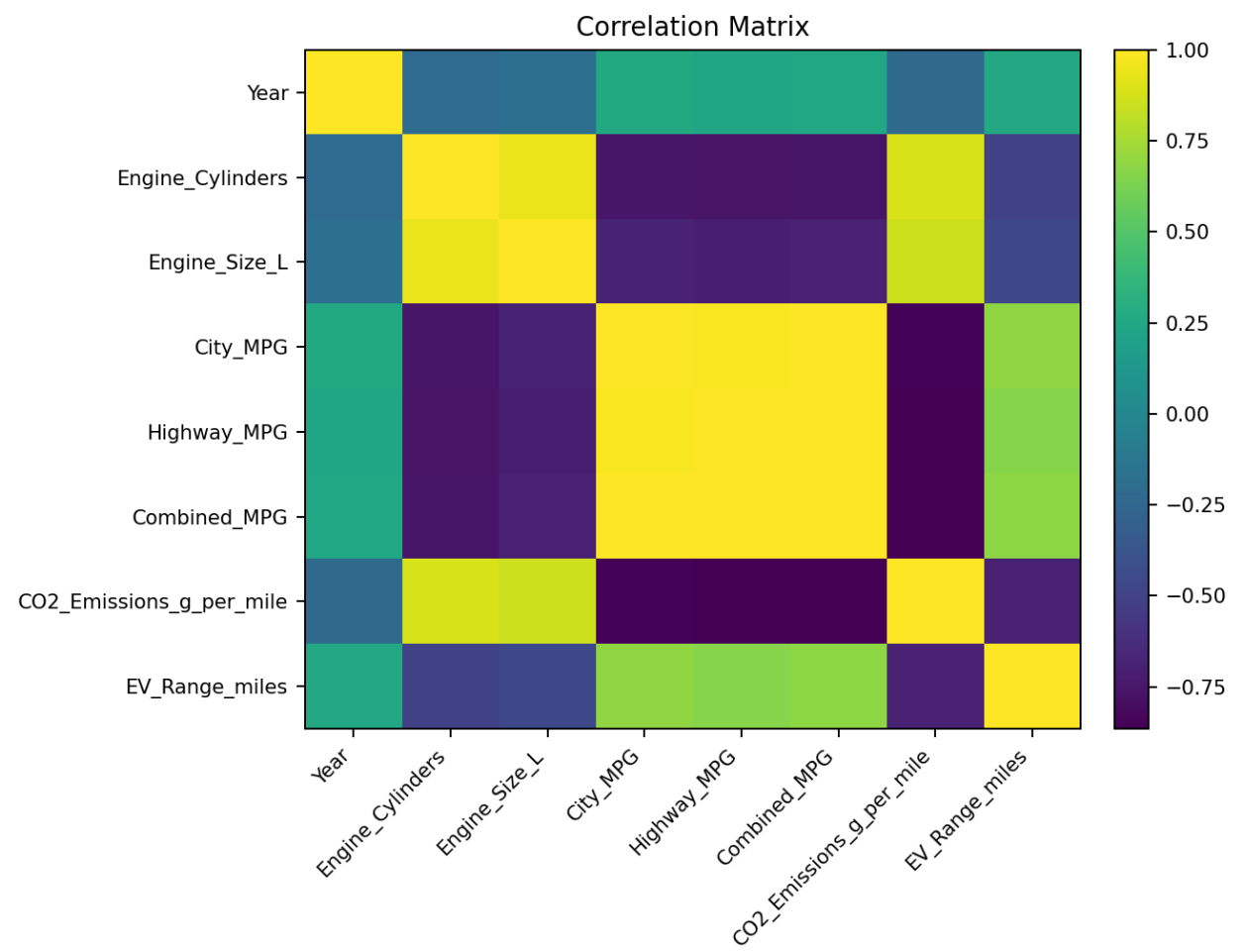


Figure 5. Correlation among numeric variables.

5. Numeric Feature Summary

Feature	Mean	Std	Min	Max
Year	2,020.47	3.41	2,015.00	2,026.00
Engine_Cylinders	5.04	2.38	0.00	16.00
Engine_Size_L	2.80	1.53	0.00	8.40
City_MPG	28.01	23.87	8.00	153.00
Highway_MPG	32.99	19.19	11.00	142.00
Combined_MPG	29.75	21.70	9.00	146.00
CO2_Emissions_g_per_mile	364.09	153.79	0.00	979.00
EV_Range_miles	37.78	108.49	0.00	640.00

6. Final Preprocessing Decisions

- Fill the missing value in **Fuel_Type** with **Unknown**.
- Remove duplicated rows using **drop_duplicates()**.
- Simplify **Transmission** into a lower-cardinality feature named **Transmission_Type**.
- Drop **Model** for a first modeling attempt because it has high cardinality.
- Scale numeric variables using **StandardScaler**.
- Encode categorical variables using **OneHotEncoder(handle_unknown="ignore")**.
- Use a **ColumnTransformer** pipeline so the workflow remains clean and reusable.

7. Conclusion

The dataset is already in good condition and requires only light preprocessing. After handling the missing value, removing duplicates, simplifying one categorical feature, and building a preprocessing pipeline, the dataset becomes suitable for modeling.

The strongest EDA findings are the dominance of ICE (Gas) vehicles, the large difference in CO₂ emissions across categories, and the distinctive EV range behavior. This makes the dataset well suited for both **vehicle category classification** and **environmental impact analysis**.